

COMP410/510

Computer Graphics

Spring 2026 Syllabus

Yücel Yemez

Office: Eng139 ☎: (0212) 338 1585

e-mail: yyemez@ku.edu.tr

Office hours: TBA

Course Description and Objectives

This course aims to introduce theoretical and practical aspects of 3D computer graphics, with an emphasis on application programming. After the course, students will be able to design and implement basic 3D interactive graphics applications. Topics include graphics systems and models; geometric representations and transformations; graphics programming; input and interaction; viewing and projections; compositing and blending; illumination and color models; ray tracing; shading; texture mapping; animation; rendering and implementation; hierarchical and object-oriented modeling; scene graphs; 3D modeling.

Course Prerequisites

Ability to write application programs in C or C++. Familiarity with data structures and matrix algebra.

Course Webpage

Follow the course web page on Learn Hub closely to catch up with reading assignments, homework, exams, projects, and any other course-related announcements.

Lectures

Tuesday & Thursday 11:30-12:40, SCI-103

Textbook

There are no mandatory textbooks, but we will mostly follow the material in the following books. Students are expected to do the reading that will be assigned from these references:

Edward Angel, *Interactive Computer Graphics: A Top-Down Approach with OpenGL*, Addison-Wesley, 6th edition.

OpenGL Programming Guide, 8th edition, The Official Guide to Learning OpenGL, The Khronos OpenGL ARB Working Group.

Topics to be covered (tentative)

- Graphics systems and models
- Graphics pipeline
- Programming with Shader-based OpenGL
- Input and interaction
- Geometric objects and transformations
- Viewing & Projection
- Shading
- Building 3D models
- Buffers & Texture mapping
- Ray tracing
- Hierarchical and object-oriented modeling; scene graphs

Grading

There will be **two written exams** during the semester. The exams will assess students' understanding of the computer graphics concepts covered in the lectures (no coding involved). Online homework assignments will be given to prepare you for the exams.

An important part of the course is the **term project**. By the end of **the first month** of the semester, each student will have chosen a project topic. Projects can be either research-oriented or applications programming-oriented, addressing one of the computer graphics problems/concepts/applications covered throughout the course. Depending on the chosen topic, students may be expected to conduct a literature review of different techniques for solving the specified problem, then implement and test one of these techniques. A software implementation is mandatory, using **C/C++** and **OpenGL**. Note that you are **not** allowed to use any other API, such as Unity, for your projects. A **progress** presentation, including a report and code submission, will be required by the end of **the second month** of the semester. Finalized projects will be submitted and presented at the end of the semester.

Final grades will be composed of:

Written exams	40%
Homework	10%
Project progress	20%
Final project	30%

All material, code, and documentation submitted for exams, assignments, and projects must be your own work.

Enjoy the course!